Ceramics International 50 (2024) 26513–26527

Contents lists available at ScienceDirect

Ceramics International

journal homepage: www.elsevier.com/locate/ceramint

Analysis of slag chemistry in WEEE smelting using experimental and modelling study of the "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system in equilibrium with Cu metal

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A R T I C L E I N F O

Handling Editor: Dr P. Vincenzini

Keywords:
Copper
WEEE
e-scrap
Recycling
Phase equilibria
Liquidus

A B S T R A C T

The variability of the composition of recycled copper-containing materials requires enhanced control and understanding of slag chemistry in the secondary pyrometallurgical processing, when compared to primary copper smelting. In the black copper route for recycling waste electronic and electrical equipment (WEEE), slags exhibit high concentrations of alumina and zinc oxide. The simplest system governing phase equilibria at the reducing smelting stage of the black copper process is "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$. In the present study, slags within this system were equilibrated with liquid copper Cu and solid Fe metallic alloys, quenched and characterized by the Electron Probe X-ray Microanalysis (EPMA). The range of compositions was selected based on the information about the process available in literature. The study focused on the liquidus and proportion of solid spinel in the temperature range from 1100 to 1300 °C. Precise control of the proportion of solids can enhance the stability of refractory materials against corrosive slags while maintaining entrained metal droplets at reasonably low level. Additionally, experimental correlations between the solubility of copper in the oxide liquid and the partial pressure of oxygen were developed for the Zn-free slags in equilibrium with metallic copper at 1200 and 1300 °C. All experimental results were compared to thermodynamic predictions using recent models and FactSage® software. Uncertainties were identified to be used in further model improvement.

1. Introduction

Recycling of Cu and other valuable elements containing residues into new products is the key to save natural resources and energy [1,2]. After physical preparation, elements from various materials are separated among different phases during metallurgical recycling processing [3]. Pyrometallurgical treatment is often used as an important step for copper-containing materials [4], with the key phases being molten slag (oxide liquid), metal or matte. In the so called "black copper" route [5,6], Cu, Ni, Sn, Pb, Ag, Au and PGM's (platinum group metals) are predominantly partitioned into a liquid copper-based metal, while less valuable elements, e.g. Ca, Si, Al and others, form a molten oxide solution slag phase (Fig. 1) [7]. The slag product also has value and can be used as a construction material, if the concentrations of heavy metals in it are maintained at low levels, determined by legislations. Materials to be recycled are usually represented by manufacturing waste or end of life consumer waste, e.g. the WEEE (waste of electric and electronic equipment) and PCB (printed circuit board) scrap. Recycling of electronic waste is widely discussed in literature, with focus on material composition [8,9], energy consumption [10,11] and thermodynamic considerations of the elemental distribution between slag and metal [12–18]. Other recycling materials used in secondary smelting can be ashes, dusts, slags, sludges and shredded materials from diverse material preprocessing operations. Inherent variability of the feed material is one of the key challenges for efficient pyrometallurgical treatment.

The ability to predict the products and energy balance based on the input materials and process parameters is critical to increase the overall stability and decrease operating costs. Recent advancements in the thermodynamic software [20,21] and the development of large self-consistent thermodynamic databases allow such predictions. They can serve as tools for the process control and optimization. Present study is a part of a broader research program supported by an international consortium of lead, zinc, copper and iron producing and recycling companies. It is focused on the development of a self-consistent set of the

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https://doi.org/10.1016/j.ceramint.2024.04.380
Received 19 February 2024; Received in revised form 9 April 2024; Accepted 26 April 2024
Available online 27 April 2024

Fig. 1. Example of recycled materials, a furnace used to process them and the slag chemistry [19]. Sources of photos: www.aurubis.com, www.christiansen-ing.de.

thermodynamic parameters for the Cu–Pb–Zn–Fe–Ca–Si–Al–Mg–O–S-(As, Sn, Sb, Bi, Ag, Au, Ni, Cr, Co and Na) system, gas/oxide liquid/-matte/speiss/metal/solids phases, using the integrated experimental and thermodynamic modelling methodology [22–24]. The internal thermodynamic database includes more than 150 sulphide, sulphate and oxide solid and liquid solutions, and compounds and more than 100 gas species. The database is constantly being reviewed and is periodically updated to achieve a self-consistent set of parameters for the solutions and compounds in the system. This then allows the accurate prediction of phase equilibria and other thermodynamic properties over the wide range of compositions, temperatures and pressures.

Phase equilibria in the "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system largely determines the operation window for slags used in the black copper route, with a typical bulk composition shown in Fig. 1. Earlier studies either described phase equilibria within the sub-systems of the selected system [25–27], or the effect of one component on the liquidus within the selected area of the system [28], provided the data on the phase equilibria in the system at the fixed component ratios (CaO/SiO_2, (CaO + SiO_2)/Al_2O_3 and others) [29–35], or focused on impurity elements [36,37]. In the present study, a compositional area relevant to the secondary copper smelting slag is selected [19] with targeted experiments focused on assessing the uncertainty of the model in the "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system. In real process, slags are in equilibrium with black copper, containing other impurities, such as Ni, Sn, Pb which are outside of this work. Very reducing conditions, close to the reduction of FeO from slag into metallic Fe, are maintained in a typical furnace to limit the partitioning of valuable metals into the slag. At the same time the presence of solid metallic iron should be avoided to improve the fluidity of slags and avoid accretions [38]. In the present study, the presence of solid iron is targeted to demonstrate the limiting conditions for reduction. The presence of metallic or oxide solids in the slag interferes with the settling of entrained metallic droplets and should be minimized. On the other hand, a certain well-controlled proportion of solids, particularly spinel, helps improving the stability of refractory bricks and oxygen lance towards the corrosive and abrasive action of slags by stabilizing the "sealing solids" layer on the surface. The "sealing solids" then form a protective layer between the slag and the refractory, limiting the dissolution and mechanical washing of the latter [39,40]. Thus, experiments were designed to obtain slags in equilibrium with metallic copper, metallic solid iron and one or more solid oxide phases.

2. Research methodology

The experimental procedure and apparatus used in the present study were described earlier by the authors [41–44]. The initial mixtures were made of high-purity Cu_2O, Al_2O_3, ZnO, Fe_2O_3 and SiO_2 (all ≥99.9 wt% purity, supplied by Alfa Aesar, MA, USA or by Sigma-Aldrich, MO, USA) oxide and Cu and Fe (≥99.9 wt% purity; supplied by Alfa Aesar, MA, USA) metallic powders. CaO was added to the mixture in a form of the "master slag" CaO–SiO_2, 4:6 M ratio, prepared in advance by high-temperature sintering of mixed $CaCO_3$ and SiO_2 powders (≥99.9 wt % purity; supplied by Alfa Aesar, MA, USA). Preliminary equilibration showed that promoting the formation of calcium silicates decreases the equilibration time required for further experiments. To ensure that metallic copper and iron were present at the final equilibrium in all samples, 10–20 wt% of excess metal was added to the oxide mixtures. About 0.5 g of mixture was used in each equilibration experiment. The initial compositions of the mixtures were selected so that one or more crystalline phases were present in equilibrium with liquid slag and metals. The initial predictions of the targeted phase compositions were made for every sample using FactSage® 8.2 software [20,21] and continuously updated internal thermodynamic database [22–24]. The volume fraction of solids in the final sample after equilibration was targeted to be in the range from 10 to 50 %, and preferably about 10 %. Low proportion of solids ensures faster equilibration time, and facilitates quenching of liquid slag into amorphous material without change in composition. Solid phases acted as heterogeneous nucleation centres, so that a minimum distance of 10–20 μm between solid grains was necessary to ensure that an amorphous slag of uniform composition, unaffected by dendritic crystal growth during quenching, was obtained. The correlation between the proportion of the solid phases present in equilibrium and the distance between solid grains was described previously by the authors [43]. An iterative procedure involving preliminary experiments was required to achieve the targeted phase proportions for a given equilibration temperature and bulk slag composition, since the exact liquidus compositions at a given temperature were not initially known. Homogenous slag areas, not affected by the growth of the "quench crystals", were analyzed and included in the present study.

The mixtures were pressed into pellets using a tool steel die before being mounted on a substrate. Alumina crucibles (1.5 cm long, 1 cm outer diameter, 99.7 wt% purity, supplied by Xiamen Wintrustek Advance Materials Co., LTD, China) were used as sample support substrates for equilibration. For the closed system experiments the alumina crucibles were additionally sealed inside silica ampoules (5 cm long, 1.3 cm outer diameter, 99.999 %, supplied by GY, Jiangsu, China) to

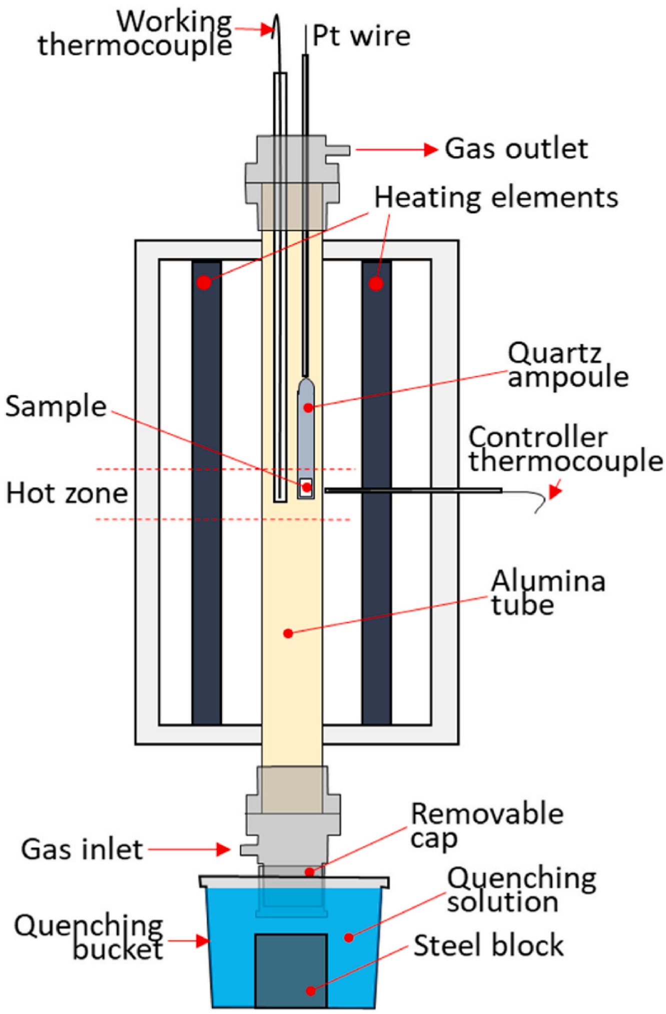

Fig. 2. Schematic of the furnace setup used in the present study for the closed system experiments.

prevent the contamination of the samples with the suspension support materials and to limit the evaporation of the volatile species from the melt. The ceramic crucibles were previously found to be reliable as substrates [43,45,46]. The precipitation of the primary phase on the ceramic substrate prevents the excessive dissolution of the substrate material into the slag ensuring little change in the bulk composition of the system during the equilibration time [47,48]. Small changes are tolerable, because the useful data are obtained by measuring the final compositions of individual phases after the equilibrium.

Each equilibration experiment was carried out in an electrical-resistance heated vertical tube furnace with an impervious recrystallized alumina (30-mm inner diameter) reaction tube with silicon carbide (SiC) heating element; the upper temperature limit for the furnace operation was 1700 °C. The samples were placed immediately adjacent to a working thermocouple (type B) that was encased in a recrystallized alumina sheath in the uniform hot zone of the furnace to monitor the actual sample temperature. The working thermocouple was calibrated against a standard thermocouple (supplied by the National Measurement Institute of Australia, NSW, Australia). The overall absolute temperature accuracy of the experiment was estimated to be ±3 K. The samples were suspended in the hot zone of the furnace by Kanthal support wire (Fe–Cr–Al alloy, 0.7 (Kanthal D) or 1-mm (Kanthal A-1) diameter; melting temperature 1500 °C; suitable for operating up to p(O_2) = 1 atm; supplied by Vulcan Stainless, VIC, Australia). The samples were then equilibrated at the final target temperature for the required time. For the open system experiments the p(O_2) was maintained by an accurate control of the CO/CO_2 ratio in the gas flow phase using calibrated U-tube capillary flowmeters. The desired flowrates of gases, corresponding to the indicated p(O_2) were calculated using FactSage 8.2® FactPS database [21] for the ideal gas phase. The accuracy of the oxygen potential was confirmed using a DS-type oxygen probe (supplied and calibrated by Australian Oxygen Fabricators, Melbourne, VIC, Australia). At the end of the equilibration process, the samples were released and rapidly quenched into $CaCl_2$ brine solution at −20 °C. Brine solution was chosen over water due to the higher cooling rates [49,50]. The schematic of the described furnace setup was given in Fig. 2. Fig. 2 demonstrates the setup for the closed system experiments; at the same time the setup is the same for the open system experiment, except for the absence of the sealed quartz ampoule protecting the sample. The samples were then washed thoroughly in water and ethanol, dried on a hot plate, mounted in epoxy resin and cross-sectioned and polished using conventional metallographic techniques.

The cross-sectioned samples were examined by optical microscopy, carbon-coated, and the compositions of the phases were measured by EPMA using wavelength-dispersive detectors (JEOL 8200L EPMA; Japan Electron Optics Ltd., Tokyo, Japan) operated at 15 kV accelerating voltage and 20 nA probe current. The standard Duncumb–Philibert atomic number, absorption, and fluorescence (ZAF) correction supplied by JEOL [51–53] was used. Copper (Cu) (for Cu and "$CuO_{0.5}$" measurements), corundum (Al_2O_3) (for $AlO_{1.5}$ measurements), hematite (Fe_2O_3) (for Fe, FeO and $FeO_{1.5}$ measurements) zincite (ZnO) and wollastonite ($CaSiO_3$) (for CaO and SiO_2 measurements) (supplied by Charles M. Taylor Co., Stanford, CA) were used as EPMA standards. The measurement uncertainties corresponded to the use of metallic copper as a standard for oxide measurements and hematite (Fe_2O_3) as a standard for metal measurements did not exceed 0.2 %. Only concentrations of metal cations were measured using EPMA; no data on the oxidation states of the metal cations in slag were obtained in this study. For presentation purposes, the copper and iron oxide concentrations were then re-calculated as $CuO_{0.5}$ and FeO, respectively (notation "$CuO_{0.5}$" and "FeO" used). The oxidation state +1 for Cu was selected according to the information on the slags in equilibrium with metal in the Cu–Si–O system [54]. It was shown [55] that the increase from fully focused electron beam (zero probe diameter) to 50 μm decreased the uncertainty of the copper slags compositions measured with EPMA and did not affect the accuracy of measurements. Thus, the probe diameter was selected depending on the size of the phase under investigation. The slag compositions reported in this study were measured mainly using 30 μm probe diameter. The compositions of the phases reported in the present study are the average values of a number (from 15 to 25, usually 20) of independent measurements from different regions of the samples rather than single measurements.

The accuracy of the compositional measurements was improved using the standard JEOL ZAF correction following an approach similar to that described previously by the authors [43,44,56–59]. This involved using stoichiometric compounds of known composition as secondary internal standards. The measured compositions of some stoichiometric compounds, $CuAlO_2$ [44], Fe_2SiO_4 [56], $CuSiO_3$ [59], $ZnAl_2O_4$ [60], Zn_2SiO_4 [61], $Ca_2Fe_2O_5$, $CaFe_2O_4$ and $CaFe_4O_7$ [62], measured by EPMA were found to be shifted from their stoichiometry, when only standard JEOL ZAF correction were used. The following equations (1-6) were then developed and applied to correct all cations concentrations in the target system of the present study:

$$x_{CuO_x}^{corrected} = x_{CuO_x}\left(1 + 0.035x_{CaO} + 0.072x_{SiO_2} + 0.116x_{AlO_{1.5}}\right), \tag{1}$$

$$x_{ZnO}^{corrected} = x_{\mathrm{CaO}}\left(1 + 0.07x_{SiO_2} + 0.06x_{AlO_{1.5}}\right), \tag{2}$$

Table 1
Experimental results for the Zn-free "$CuO_{0.5}$"-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system in equilibrium with Cu metal.

#	Substrate	Temperature, °C		Time, hours	Phase	Compositions, wt.%					Ox./ *Me.*	log $p(O_2)$, atm
						"$CuO_{0.5}$"	"FeO"	CaO	SiO_2	$AlO_{1.5}$	Ox	
		Exp.	Pred.			*Cu*	*Fe*	*Ca*	*Si*	*Al*	*Me*	
oxide liquid + spinel + Cu metal at fixed $p(O_2)$												
1	Al_2O_3	1300	1338	5	oxide liquid	1.4 ± 0.0	28.8 ± 0.1	6.3 ± 0.0	38.8 ± 0.1	24.7 ± 0.1	Ox	−9[a]
					spinel	0.13 ± 0.05	42.3 ± 0.7	0.06 ± 0.03	0.07 ± 0.01	57.4 ± 0.7	Ox	
					Cu metal	***99.85 ± 0.03***	***0.14 ± 0.02***	***0.01 ± 0.01***	***0 ± 0.01***	***0.00 ± 0.00***	***Me***	
2	Al_2O_3	1300	1333	17	oxide liquid	1.4 ± 0.1	25.7 ± 0.1	6.6 ± 0.0	40.4 ± 0.1	25.9 ± 0.1	Ox	−9[a]
					spinel	0.19 ± 0.01	42 ± 0.3	0.08 ± 0.02	0.1 ± 0.03	57.6 ± 0.3	Ox	
					Cu metal	***99.81 ± 0.02***	***0.14 ± 0.02***	***0.01 ± 0.01***	***0.02 ± 0.02***	***0.01 ± 0.01***	***Me***	
3	Al_2O_3	1300	1334	72	oxide liquid	0.72 ± 0.05	25 ± 0.1	6.7 ± 0.0	41 ± 0.1	26.6 ± 0.1	Ox	−10[a]
					spinel	0.04 ± 0.04	40.8 ± 0.1	0.04 ± 0.01	0.04 ± 0.02	59.1 ± 0.1	Ox	
					Cu metal	***99.43 ± 0.07***	***0.43 ± 0.04***	***0.01 ± 0.00***	***0.03 ± 0.02***	***0.00 ± 0.00***	***Me***	
4	Al_2O_3	1300	1335	17	oxide liquid	0.76 ± 0.05	27.1 ± 0.1	6.5 ± 0.0	40 ± 0.1	25.6 ± 0.1	Ox	−10[a]
					spinel	0.13 ± 0.00	41.6 ± 0.0	0.02 ± 0.00	0.06 ± 0.00	58.2 ± 0.0	Ox	
					Cu metal	***99.51 ± 0.07***	***0.37 ± 0.04***	***0.01 ± 0.00***	***0.02 ± 0.02***	***0.00 ± 0.00***	***Me***	
5	Al_2O_3	1200	1241	17	oxide liquid	1.5 ± 0.1	46.7 ± 0.1	6.4 ± 0.0	33.4 ± 0.1	12 ± 0.1	Ox	−9[a]
					spinel	0.05 ± 0.02	65.6 ± 0.4	0.26 ± 0.32	0.21 ± 0.01	33.8 ± 0.2	Ox	
					Cu metal	***99.95 ± 0.04***	***0.04 ± 0.03***	***0.00 ± 0.00***	***0.01 ± 0.02***	***0.00 ± 0.00***	***Me***	
6	Al_2O_3	1200	1247	17	oxide liquid	0.77 ± 0.06	53.2 ± 0.2	6.9 ± 0.0	29.2 ± 0.1	9.9 ± 0.1	Ox	−10[a]
					spinel	0.04 ± 0.04	57.8 ± 0.8	0.08 ± 0.03	0.12 ± 0.02	41.9 ± 0.8	Ox	
					Cu metal	***99.73 ± 0.03***	***0.24 ± 0.01***	***0.01 ± 0.01***	***0.01 ± 0.01***	***0.01 ± 0.01***	***Me***	
7	Al_2O_3	1200	1249	17	oxide liquid	0.81 ± 0.05	52.9 ± 0.2	6.5 ± 0.0	29.6 ± 0.1	10.2 ± 0.1	Ox	−10[a]
					spinel	0.04 ± 0.03	57.9 ± 0.1	0.12 ± 0.03	0.21 ± 0.06.	41.7 ± 0.1	Ox	
					Cu metal	***99.78 ± 0.04***	***0.19 ± 0.02***	***0.01 ± 0.01***	***0.01 ± 0.01***	***0.01 ± 0.02***	***Me***	

[a] $p(O_2)$ over the system was controlled during the experiment.

$$x_{FeO_x}^{corrected} = x_{FeO_x}\Big(1 + 0.0208x_{CaO} - 0.0119x_{CaO}^2 + +0.0119x_{FeO_x}\,x_{CaO} + 0.0637x_{SiO_2} - 0.0637x_{SiO_2}^2 + 0.038x_{AlO_{1.5}}\Big), \quad (3)$$

$$x_{CaO}^{corrected} = x_{\mathrm{CaO}}\Big(1 - 0.0208x_{FeO_x} - 0.0119x_{FeO_x}^2 + 0.0119x_{CaO}x_{FeO_x} - 0.044x_{AlO_{1.5}} - 0.035x_{CuO_x}\Big), \quad (4)$$

$$x_{SiO_2}^{corrected} = x_{SiO_2}(1 - 0.07x_{ZnO} - 0.0637x_{FeO_x} + 0.0637x_{SiO_2}x_{FeO_x} - 0.072x_{CuO_x}), \quad (5)$$

$$x_{AlO_{1.5}}^{corrected} = x_{AlO_{1.5}}(1 - 0.06x_{ZnO} - 0.038x_{FeO_x} + 0.044x_{\mathrm{CaO}} - 0.116x_{CuO_x}), \quad (6)$$

where x(CuO_x, ZnO, FeO_x, CaO, SiO_2, $AlO_{1.5}$) are the cation molar fractions after the standard JEOL ZAF corrections, and x(CuO_x, ZnO, FeO_x, CaO, SiO_2, $AlO_{1.5}$)^{corrected} are the corrected cation molar fractions respectively.

To ensure the achievement of equilibrium in the samples, the "4-point test" [42,63] was used including: (1) variation of equilibration time; (2) assessment of the compositional homogeneity of the phases by EPMA; (3) approaching the final equilibrium point from different starting conditions; and, importantly, (4) consideration of reactions specific to this system that may affect the achievement of equilibrium or reduce the accuracy. Simultaneous consideration and application of all 4 "points" of the discussed test for every experiment allows to be sufficiently confident that the phases present in those samples that passed the test are in equilibrium and are suitable to be used in the further study. The conditions of interest for the present study were characterized by the interval of temperatures and oxygen partial pressure $p(O_2)$ representing the industrial conditions for Cu recycling. At the same time the presence of Zn in the system made it impossible to run the equilibration experiments in the open system at the controlled $p(O_2)$. The partial pressure of zinc (p(Zn)) is high at high temperature and reducing conditions and increases with increasing temperature and decreasing $p(O_2)$ [64,65]. The open system equilibration approach leads to the continuous evaporation of zinc from the condensed phases into the gaseous phase, according to the following reactions [65]:

$$ZnO_{slag} \leftrightarrow Zn_{gas} + O_{slag} \quad (7)$$

$$ZnO_{slag} + CO_{gas} \leftrightarrow Zn_{gas} + CO_{2\,gas} \quad (8)$$

$$2FeO_{slag} + ZnO_{slag} \leftrightarrow Fe_2O_{3\,slag} + Zn_{gas} \quad (9)$$

The discussed reactions, and reaction (8) in particular, creates a kinetic barrier between the slag surface and the CO/CO_2 gas flow, making open system equilibration for the Zn containing systems at reducing conditions practically impossible. To overcome this issue, the experimental methodology similar to the one described before by the authors [65] was developed and used in the present study. The methodology involves closed system experiments in the sealed ampoules with the introduction of the Cu_{metal}/Cu_2O_{slag} couple to be used as internal measure of the effective $p(O_2)$. In the presence of metallic Cu, the change in the oxidation potential of the system results in the change of the solubility of "$CuO_{0.5}$" in the slag phase, according to the following reactions [65]:

$$2Cu_{metal} + O_{slag} \leftrightarrow Cu_2O_{slag} \quad (10)$$

$$2Cu_{metal} + Fe_2O_{3\,slag} \leftrightarrow Cu_2O_{slag} + 2FeO_{slag} \quad (11)$$

The Cu_{metal}/Cu_2O_{slag} couple then reflects the effective $p(O_2)$ of the slag phase as per the following reaction:

$$2Cu_{metal} + \frac{1}{2}O_{2\,gas} \leftrightarrow Cu_2O_{slag} \quad (12)$$

According to reaction (12), it is possible to estimate the effective $p(O_2)$ over the system based on the concentration of "$CuO_{0.5}$" in the oxide liquid, which is in equilibrium with Cu metal. The correlation between the $p(O_2)$ in the system and the concentration of $CuO_{0.5}$ in the oxide liquid of the system of interest is required. In the present study the following approach was used:

1) Establishing the correlation between "$CuO_{0.5}$"in oxide liquid and $p(O_2)$ over the system for the zinc-free "$CuO_{0.5}$"-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system in equilibrium with Cu metal using accurate

Table 2
Experimental results for the "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system in equilibrium with liquid Cu and solid Fe alloys.

#	Substrate	Temperature, °C		Time, hours	Phase	Compositions, wt.%						Ox./*Me.*	log p(O_2), atm
						"$CuO_{0.5}$"	ZnO	"FeO"	CaO	SiO_2	$AlO_{1.5}$	Ox	
		Exp.	Pred.			*Cu*	*Zn*	*Fe*	*Ca*	*Si*	*Al*	*Me*	
oxide liquid + spinel + Cu alloy + Fe alloy													
8	Al_2O_3	1320	1321	5	oxide liquid	0.30 ± 0.07	2.2 ± 0.1	49.7 ± 0.3	6.1 ± 0.1	28.2 ± 0.2	13.6 ± 0.4	Ox	−10.9[b]
					spinel	0.05 ± 0.04	9.0 ± 0.3	34.6 ± 0.5	0.08 ± 0.02	0.09 ± 0.06	56.1 ± 0.3	Ox	
					Cu alloy	***92.7 ± 0.3***	***4.0 ± 0.1***	***3.3 ± 0.4***	***0.01 ± 0.00***	***0.00 ± 0.00***	***0.00 ± 0.00***	***Me***	
					Fe alloy	***10.0 ± 0.7***	***0.25 ± 0.08***	***89.4 ± 0.9***	***0.06 ± 0.2***	***0.18 ± 0.6***	***0.07 ± 0.21***	***Me***	
9	Al_2O_3	1220	1258	5	oxide liquid	0.21 ± 0.07	2.7 ± 0.2	59.4 ± 1.6	5.8 ± 0.3	24.3 ± 1.1	7.6 ± 0.4	Ox	−11.4[b]
					spinel	0.02 ± 0.03	11.6 ± 0.8	35.5 ± 0.6	0.11 ± 0.02	0.18 ± 0.09	52.6 ± 0.3	Ox	
					Cu alloy	***93.4 ± 0.7***	***3.4 ± 0.5***	***3.2 ± 0.8***	***0.02 ± 0.03***	***0.02 ± 0.03***	***0.01 ± 0.01***	***Me***	
					Fe alloy	***9.2 ± 0.1***	***0.23 ± 0.09***	***90.6 ± 0.2***	***0.03 ± 0.03***	***0.00 ± 0.00***	***0.02 ± 0.01***	***Me***	
10	Al_2O_3	1220	1237	5	oxide liquid	0.18 ± 0.06	6.1 ± 0.1	54.4 ± 0.4	5.8 ± 0.1	26.4 ± 0.4	7.1 ± 0.2	Ox	−11.5[b]
					spinel	0.02 ± 0.03	21.5 ± 0.7	24.4 ± 0.8	0.10 ± 0.03	0.23 ± 0.24	53.8 ± 0.1	Ox	
					Cu alloy	***88.9 ± 0.9***	***7.4 ± 0.1***	***3.7 ± 0.9***	***0.02 ± 0.02***	***0.01 ± 0.01***	***0.01 ± 0.01***	***Me***	
					Fe alloy	***8.2 ± 0.2***	***0.43 ± 0.05***	***91.3 ± 0.2***	***0.02 ± 0.02***	***0.01 ± 0.01***	***0.01 ± 0.01***	***Me***	
11	Al_2O_3	1220	1236	5	oxide liquid	0.2.0 ± 0.07	3.3 ± 0.1	54.6 ± 0.2	6.2 ± 0.1	27.6 ± 0.1	8.1 ± 0.1	Ox	−11.5[b]
					spinel	0.04 ± 0.06	14.6 ± 1	30.9 ± 1.4	0.15 ± 0.04	0.36 ± 0.15	53.9 ± 0.5	Ox	
					Cu alloy	***92.3 ± 0.5***	***4.4 ± 0.1***	***3.3 ± 0.6***	***0.00 ± 0.01***	***0.01 ± 0.01***	***0.02 ± 0.03***	***Me***	
					Fe alloy	***7.7 ± 1.6***	***0.27 ± 0.01***	***92.0 ± 1.6***	***0.00 ± 0.00***	***0.00 ± 0.00***	***0.01 ± 0.01***	***Me***	
12	Al_2O_3	1170	1201	17	oxide liquid	0.15 ± 0.07	3.0 ± 0.1	54.1 ± 0.2	6.6 ± 0.1	28.9 ± 0.2	7.3 ± 0.1	Ox	−12.0[b]
					spinel	0.01 ± 0.01	13.6 ± 0.3	32.5 ± 0.6	0.18 ± 0.07	0.42 ± 0.27	53.3 ± 0.5	Ox	
					Cu alloy	***92.8 ± 0.4***	***3.4 ± 0.2***	***3.8 ± 0.5***	***0.01 ± 0.01***	***0.00 ± 0.00***	***0.01 ± 0.01***	***Me***	
					Fe alloy	***7.6 ± 0.8***	***0.15 ± 0.06***	***92.2 ± 0.8***	***0.01 ± 0.02***	***0.01 ± 0.01***	***0.00 ± 0.01***	***Me***	
13	Al_2O_3	1120	1166	17	oxide liquid	0.13 ± 0.04	1.8 ± 0.1	44.6± 0.2	5.6 ± 0.0	36.8 ± 0.1	11.1 ± 0.1	Ox	−12.8[b]
					spinel	0.04 ± 0.02	14.2 ± 0.3	26.8 ± 3.3	0.03 ± 0.00	0.14 ± 0.07	58.8 ± 2.9	Ox	
					Cu alloy	***95.2 ± 0.3***	***2.5 ± 0.2***	***2.3 ± 0.3***	***0.03 ± 0.05***	***0.00 ± 0.01***	***0.01 ± 0.02***	***Me***	
					Fe alloy	***7.8 ± 0.4***	***0.06 ± 0.05***	***92.1 ± 0.4***	***0.00 ± 0.00***	***0.00 ± 0.01***	***0.02 ± 0.02***	***Me***	
oxide liquid + spinel + tridymite + willemite + Cu alloy + Fe alloy													
14	SiO_2	1150	1180	168	oxide liquid	0.12 ± 0.09	21.6 ± 0.2	18.9 ± 0.3	5.3 ± 0.0	44.7 ± 0.2	9.5 ± 0.1	Ox	NA
					spinel	0.08 ± 0.09	39.6 ± 0.2	4.1 ± 0.2	0.1 ± 0.03	0.44 ± 0.32	55.7 ± 0.4	Ox	
					tridymite[a]	0.03 ± 0.04	0.96 ± 0.21	0.6 ± 0.21	0.11 ± 0.07	98.1 ± 0.6	0.19 ± 0.16	Ox	
					willemite	0.07 ± 0.03	57.7 ± 0.2	15.1 ± 0.0	0.04 ± 0.01	27 ± 0.2	0.11 ± 0.02	Ox	
					Cu alloy	***70.2 ± 0.7***	***27.7 ± 0.8***	***2.0 ± 0.2***	***0.01 ± 0.01***	***0.00 ± 0.00***	***0.01 ± 0.02***	***Me***	
					Fe alloy	***4.9 ± 0.0***	***1.4 ± 0.1***	***93.6 ± 0.1***	***0.02 ± 0.01***	***0.00 ± 0.00***	***0.02 ± 0.01***	***Me***	
oxide liquid + feldspar + tridymite + willemite + Cu alloy + Fe alloy													
15	SiO_2	1100	1123	17	oxide liquid	0.09 ± 0.05	17.8 ± 0.2	14.9 ± 0.3	10 ± 0.1	47.5 ± 0.2	9.8 ± 0.2	Ox	NA
					feldspar	0.09 ± 0.00	4.3 ± 0.0	1.2 ± 0.0	17.8 ± 0.0	44.0 ± 0.0	32.5 ± 0.0	Ox	
					tridymite[a]	0.12 ± 0.10	0.94 ± 0.06	0.43 ± 0.23	0.29 ± 0.24	97.9 ± 0.9	0.27 ± 0.23	Ox	

(*continued on next page*)

Table 2 (*continued*)

#	Substrate	Temperature, °C		Time, hours	Phase	Compositions, wt.%						Ox./ *Me.*	log $p(O_2)$, atm
						"$CuO_{0.5}$"	ZnO	"FeO"	CaO	SiO_2	$AlO_{1.5}$	Ox	
		Exp.	Pred.			*Cu*	*Zn*	*Fe*	*Ca*	*Si*	*Al*	*Me*	
					willemite	0.05 ± 0.04	59.1 ± 0.2	13 ± 0.1	0.25 ± 0.1	27.5 ± 0.1	0.08 ± 0.04	Ox	
					Cu alloy	***74.6 ± 1.3***	***23 ± 1.4***	***2.2 ± 0.2***	***0.17 ± 0.1***	***0.00 ± 0.00***	***0.00 ± 0.00***	***Me***	
					Fe alloy	***4.3 ± 0.4***	***1.5 ± 0.1***	***94.1 ± 0.5***	***0.1 ± 0.03***	***0.00 ± 0.00***	***0.00 ± 0.01***	***Me***	
16	SiO_2	1090	1095	17	oxide liquid	0.09 ± 0.03	14.8 ± 0.1	19.9 ± 0.2	8.8 ± 0.0	46.9 ± 0.2	9.5 ± 0.0	Ox	NA
					feldspar	0.23 ± 0.00	1.9 ± 0.0	1.5 ± 0.0	18.9 ± 0.0	45.0 ± 0.0	32.5 ± 0.0	Ox	
					tridymite[a]	0.00 ± 0.00	0.09 ± 0.00	0.53 ± 0.00	0.17 ± 0.00	99.01 ± 0	0.2 ± 0.0	Ox	
					willemite	0.00 ± 0.00	53.5 ± 0.0	17.6 ± 0.0	0.15 ± 0.00	28.6 ± 0.0	0.09 ± 0.00	Ox	
					Cu alloy	***81.7 ± 0.0***	***16.4 ± 0.0***	***1.8 ± 0.0***	***0.03 ± 0.00***	***0.01 ± 0.0***	***0.01 ± 0.00***	***Me***	
					Fe alloy	***4.9 ± 0.0***	***0.87 ± 0.00***	***94.2 ± 0.0***	***0.00 ± 0.00***	***0.00 ± 0.00***	***0.00 ± 0.00***	***Me***	
oxide liquid + feldspar + tridymite + Cu alloy + Fe alloy													
17	SiO_2	1115	1121	17	oxide liquid	0.06 ± 0.00	13.5 ± 0.2	19 ± 0.3	8.8 ± 0.1	48 ± 0.2	10.6 ± 0.3	Ox	NA
					feldspar	0.08 ± 0.00	1.9 ± 0.0	1.2 ± 0.0	18.5 ± 0.0	44.8 ± 0.0	33.5 ± 0.00	Ox	
					tridymite[a]	0.06 ± 0.00	0.12 ± 0.00	0.39 ± 0.00	0.16 ± 0.00	99.04 ± 0.00	0.23 ± 0.00	Ox	
					Cu alloy	***78.7 ± 0.0***	***18.5 ± 0.0***	***2.4 ± 0.0***	***0.32 ± 0.00***	***0.00 ± 0.00***	***0.02 ± 0.00***	***Me***	
					Fe alloy	***6.2 ± 0.0***	***1.0 ± 0.0***	***92.7 ± 0.0***	***0.00 ± 0.00***	***0.01 ± 0.00***	***0.00 ± 0.00***	***Me***	
oxide liquid + tridymite + willemite + Cu alloy + Fe alloy													
18	SiO_2	1200	1291	17	oxide liquid	0.11 ± 0.05	22.2 ± 0.8	39.6 ± 0.8	0.02 ± 0.01	38.0 ± 0.2	0.1 ± 0.02	Ox	NA
					tridymite[a]	0.03 ± 0.05	0.52 ± 0.09	0.63 ± 0.06	0.02 ± 0.02	98.8 ± 0.1	0.01 ± 0.01	Ox	
					willemite	0.06 ± 0.04	47.5 ± 0.1	24.3 ± 0.1	0.01 ± 0.01	28.1 ± 0.0	0.00 ± 0.01	Ox	
					Cu alloy	***78.3 ± 0.2***	***20.2 ± 0.0***	***1.4 ± 0.2***	***0.02 ± 0.01***	***0.01 ± 0.01***	***0.00 ± 0.01***	***Me***	
					Fe alloy	***6.2 ± 0.4***	***1.2 ± 0.6***	***92.6 ± 0.3***	***0.01 ± 0.02***	***0.01 ± 0.01***	***0.00 ± 0.00***	***Me***	
19	SiO_2	1200	1249	17	oxide liquid	0.10 ± 0.04	23.5 ± 0.2	28.2 ± 0.2	6.3 ± 0.1	41.8 ± 0.2	0.07 ± 0.02	Ox	NA
					tridymite[a]	0.03 ± 0.05	0.52 ± 0.09	0.63 ± 0.06	0.02 ± 0.02	98.8 ± 0.1	0.01 ±0.01	Ox	
					willemite	0.00 ± 0.01	54 ± 0.1	18.2 ± 0.2	0.13 ± 0.04	27.7 ± 0.1	0.00 ± 0.00	Ox	
					Cu alloy	***72.9 ± 0.3***	***24.1 ± 0.1***	***2.9 ± 0.3***	***0.03 ± 0.01***	***0.01 ± 0.01***	***0.00 ± 0.01***	***Me***	
					Fe alloy	***5.1 ± 1.0***	***1.5 ± 0.3***	***93.4 ± 0.8***	***0.01 ± 0.02***	***0.01 ± 0.01***	***0.00 ± 0.01***	***Me***	

[a] the measurement of the phase composition was affected by secondary X-ray fluorescence and close presence of other phases.
[b] $p(O_2)$ over the system was estimated using thermodynamic model and the established correction (eq. (13)).

experimental data obtained with the open-system gas equilibration technique in the composition area of interest.

2) Extrapolating the correlation to the zinc-containing system based on the trends predicted by FactSage software [21] using internal thermodynamic database [22–24] and obtained experimental data.
3) Performing closed-system equilibration experiments for the "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ chemical system with the addition of Cu metal in sealed ampoules.
4) Deriving the effective $p(O_2)$ over the system from the FactSage software [21] using internal thermodynamic database [22–24] and adjusting the prediction based on the experimental information obtained in the steps 1–3.

3. Results and discussion

Preliminary experiments were carried out to find the conditions of achievement of equilibrium using the "4-point test" [42,63] including: (1) variation of equilibration time; (2) assessment of the compositional homogeneity of the phases by EPMA; (3) approaching the final equilibrium point from different starting conditions; and, importantly, (4) consideration of reactions specific to this system that may affect the achievement of equilibrium or reduce the accuracy. For open system experiments, equilibration time varied from 5 to 72 h to ensure that no further reactions between the condensed and gaseous phases take place in the sample. Homogeneity of the phases was examined in different locations of the samples. When compositional trends were observed, results were rejected, and equilibration time was increased. The dissolution of the substrate material into the liquid which could have

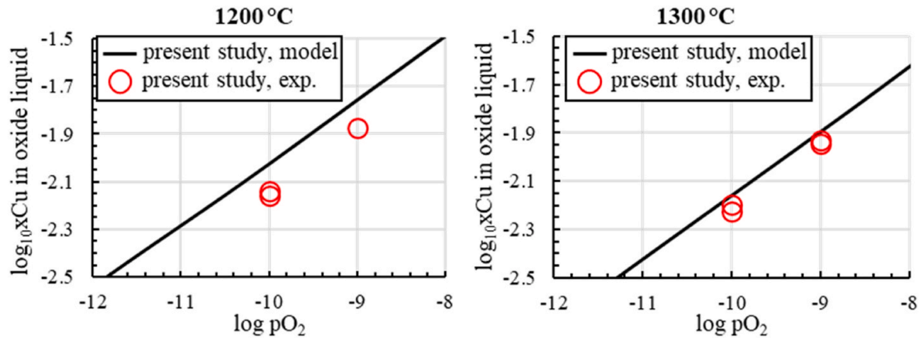

Fig. 3. Comparison of the experimental correlation between the solubility of Cu in the oxide liquid of the "$CuO_{0.5}$"-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system in equilibrium with Cu metal with the corresponding thermodynamic predictions [22–24]. In Y-axis, "xCu" is the molar ratio "$CuO_{0.5}$"/("$CuO_{0.5}$"+ FeO + $FeO_{1.5}$+ CaO + SiO_2 + $AlO_{1.5}$).

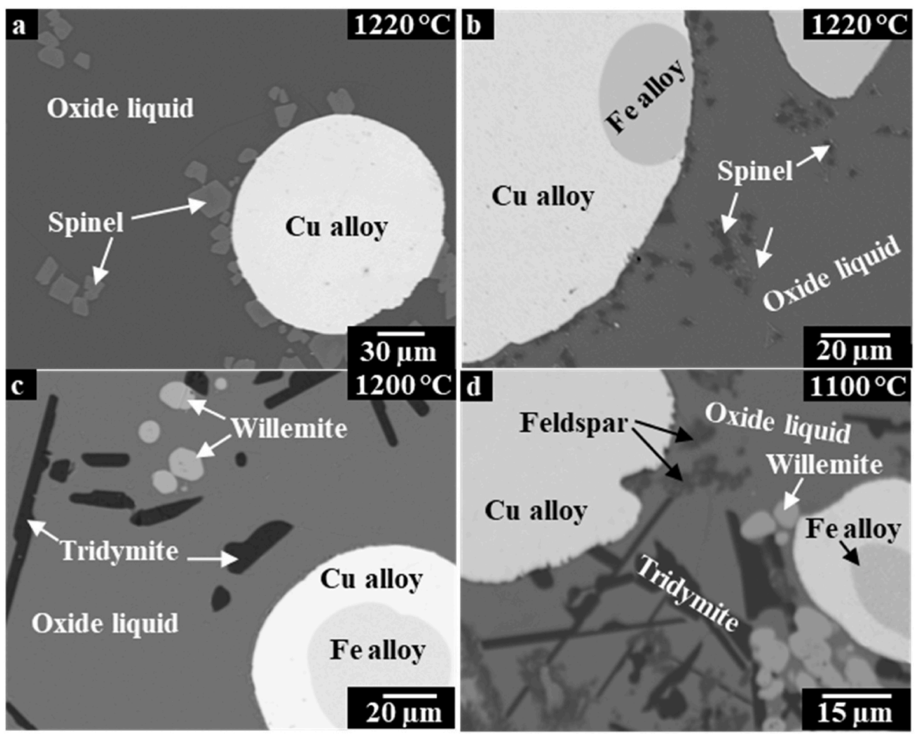

Fig. 4. Examples of the backscattered SEM micrographs of the quenched "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ samples in equilibrium with Cu or Cu/Fe alloys: (a) oxide liquid-spinel-Cu alloy; (b) oxide liquid-spinel-Cu alloy-Fe alloy; (c) oxide liquid-tridymite-willemite-Cu alloy-Fe alloy; and (d) oxide liquid-feldspar-tridymite-willemite-Cu alloy-Fe alloy.

potentially affected phase equilibria in the system was found to be not a problem, because no compositional trends were observed for the samples after equilibration and the targeted phases were obtained. Final equilibration times for the open system experiments are given in Table 1. For the closed system experiments, it was found that 5 h equilibration time was sufficient for the achievement of equilibrium at temperatures above 1200 °C, and 17 h were necessary for lower temperatures in the range from 1090 to 1200 °C. Equilibration times are listed in Table 2. Different starting compositions were used, the measured compositions of liquids agreed within the standard deviation of measurements.

The results of the experiments in the "$CuO_{0.5}$"-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system in equilibrium with Cu metal and controlled gas atmosphere are shown in Table 1. These were used to establish the correlation between $p(O_2)$ and the concentration of $CuO_{0.5}$ in oxide liquid. Solid spinel was present as well, as typically observed in WEEE recycling. The established correlation is presented and compared to the thermodynamic predictions of the current thermodynamic model [22–24] in Fig. 3.

Based on the comparison of the experimental results with the existing thermodynamic predictions, the following correction was introduced to the predicted $p(O_2)$ over the system:

$$log\ (pO_2)_{corrected} = log\ (pO_2)_{predicted} - 0.00306T + 4.07072 \quad (13)$$

where pressure is given in atmospheres, temperature in °C.

The correction (13) was applied to FactSage® 8.2 calculations for the closed-system experiments in equilibrium with metallic iron to estimate the effective $p(O_2)$.

Examples of typical microstructures observed in the studied samples of the "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system are given in Fig. 4. These micrographs were obtained using backscattered electron microscopy (BSE) to provide compositional contrast based on the mean atomic number. Oxide liquid (slag), solid phases, liquid copper metal and solid iron are clearly visible. Solid phases include spinel (Fig. 4a and b), tridymite (Fig. 4c and d), willemite (Fig. 4c and d) and feldspar

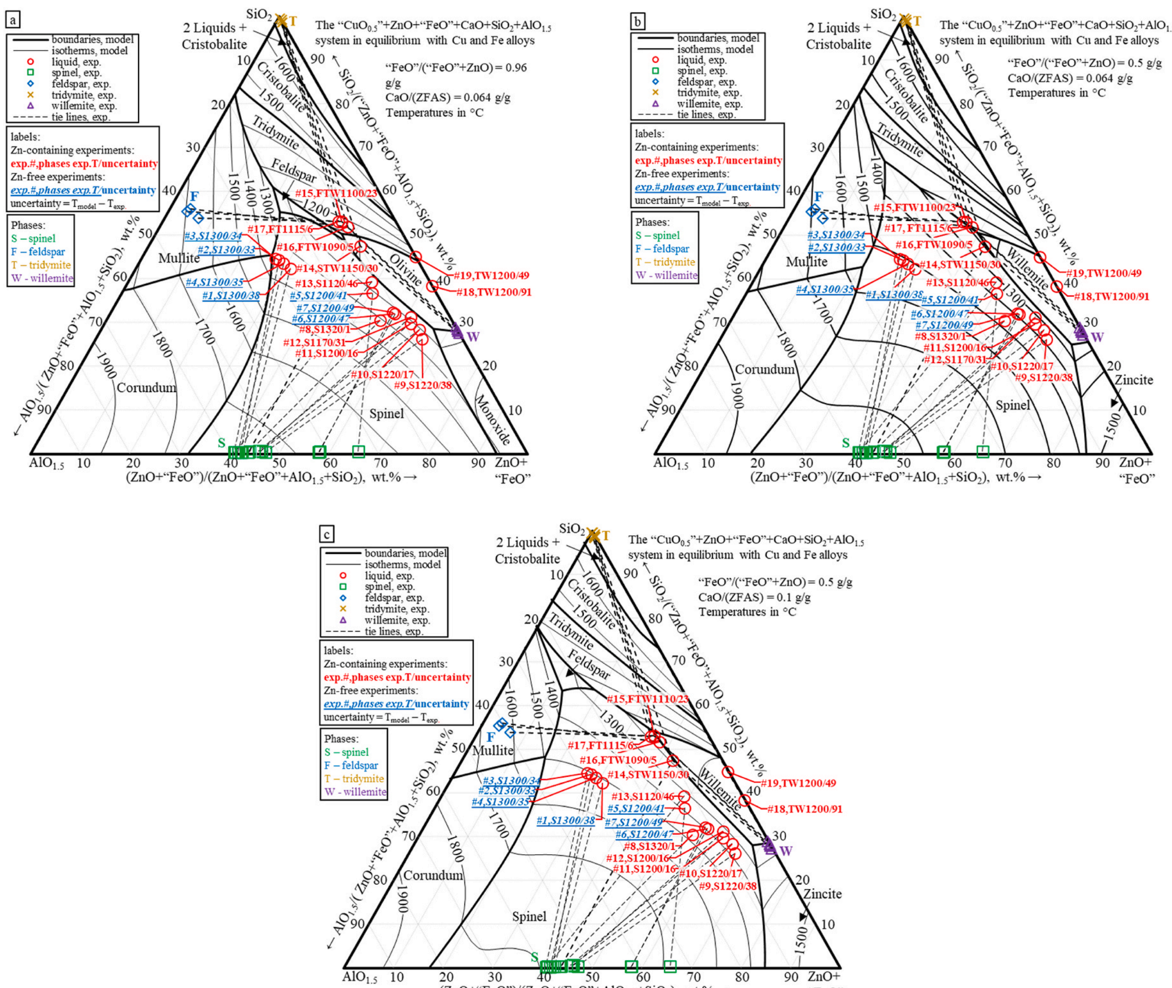

Fig. 5. Predicted liquidus surface of the "$CuO_{0.5}$"-ZnO-"FeO"-CaO-SiO_2-$AlO_{1.5}$ system in equilibrium with Cu and Fe alloys: a) "FeO"/("FeO"+ZnO) = 0.96 g/g, CaO/(ZnO+"FeO"+SiO_2+$AlO_{1.5}$) = 0.064 g/g, b) "FeO"/("FeO"+ZnO) = 0.5 g/g, CaO/(ZnO+"FeO"+SiO_2+$AlO_{1.5}$) = 0.064 g/g, and c) "FeO"/("FeO"+ZnO) = 0.5 g/g, CaO/(ZnO+"FeO"+SiO_2+$AlO_{1.5}$) = 0.1 g/g. Temperatures are in °C.

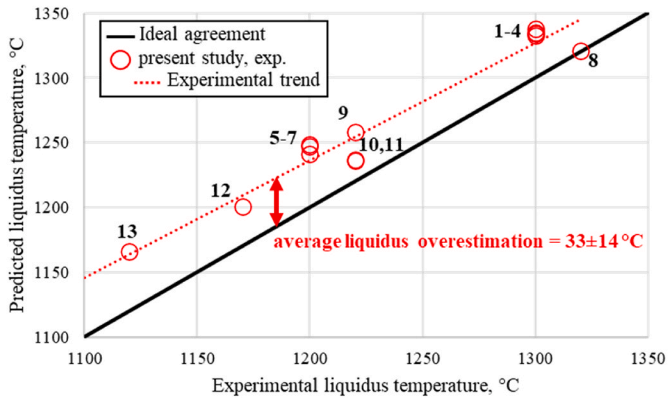

Fig. 6. Comparison of the predicted liquidus temperatures with the experimental results for the spinel primary phase in the "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system in equilibrium with Cu metal or Cu and Fe alloys.

(Fig. 4d). The average compositions of the phases obtained in the present study are reported in Table 2. The partial pressures of oxygen, $p(O_2)$ estimated using FactSage software [21], internal thermodynamic database [22–24], and corrected using Equation (13) are also provided in Table 2. The experimental temperature is provided, as well as predicted liquidus temperature for the liquid slag of the same compositions. The methodology of predictions and agreement is discussed further.

3.1. Summary of samples

The compositions of the oxide liquid (slag) for the samples of industrial interest are projected onto a (ZnO + FeO + $FeO_{1.5}$) – $AlO_{1.5}$ – SiO_2 triangle in Fig. 5a–c for the presentation purposes. The liquidus surface and primary phase fields calculated by FactSage® 8.2 are superimposed. These calculations correspond approximately to the experimental conditions. The exact comparison of model predictions and experiments for the system of so many components is difficult, and generally cannot be done using a single section through the multicomponent system. A method for comparison is discussed further. Three sections are used, which cover all primary regions of solid phase formation observed in the experiments. For relatively low concentrations of zinc oxide, corresponding to Fig. 5a, the primary phase fields are spinel, feldspar and olivine. For higher concentration of zinc oxide corresponding to Fig. 5b and c, the primary phase fields are spinel, tridymite, feldspar and willemite. Zn-free experiments are also plotted in Fig. 5, showing the tie-lines with spinel, which are noticeably different from the rest of the experiments. This difference is explained by the absence of ZnO in the system and higher $p(O_2)$ in those experiments, compared to the data-series in equilibrium with metallic Fe. The experiments cover wide range of the spinel stability for different (ZnO + "FeO")/SiO_2 ratios at temperatures from 1120 to 1320 °C, which gives more confidence in the estimating systematic deviations of thermodynamic model predictions in the area of interest. The intention is to use the predictions with corrected models at any $p(O_2)$, concentration of ZnO, CaO, $AlO_{1.5}$, etc. providing the recommendations for the specific fluxing for a given feed. These recommendations can be used as a feed-forward control tool, as well as back-feed control, and provide guidance for shifting the slag chemistry.

3.2. Comparison of multicomponent experiments and thermodynamic predictions

Two types of comparison between experiments and model predictions have been performed, both using the Equilib module of FactSage®. Fig. 6 provides the comparison in terms of liquidus temperatures at given bulk composition, corresponding to the liquid slag in the experiment. The results are for spinel primary phase. For a given composition of liquid slag in equilibrium with copper and solid iron, the perfect model would provide the temperature of the precipitation of the first solid to be the same as the experimental temperature. In this type of calculations, the composition and proportion of solid oxides were not included in the bulk input of Equilib module. The input was liquid slag of 100 g, 5 g or extra Cu and 5 g of extra Fe. Calculations with target on spinel formation have been performed with search on temperature. The amount of oxygen was adjusted iteratively to result in the final amount of slag, metallic copper and metallic iron to be within 0.1 g of the initial input. The calculated temperatures are compared to experiment. Fig. 6 shows moderate, but systematic deviation of predicted liquidus of spinel in the studied region, on average 33 ± 14 °C higher than in the experiment. Predicted liquidus temperatures for other primary phases are given in Table 2. They also show positive deviation from experiment. These points correspond to the limiting fluxing conditions.

The second type of calculation is focused on the composition and proportion of phases at the desired (lower) temperature of operation. It is practically important for the operation, which is designed to optimize the energy balance and to preserve the refractory bricks [66]. Experimentally measured phases present in the samples were combined for the calculation at fixed target proportions to form "mean bulk" compositions. These served as an input for the Equilib module. Then, thermodynamic equilibria for these "mean bulk" compositions were calculated at the temperatures and $p(O_2)$'s of the corresponding experiments. This method allows estimation of the difference between the experiments and the corresponding thermodynamic predictions in terms of the phase assemblage, elemental composition of the present phases (including solid oxides), and phase proportions. The results of the comparison are given in Table 3 for the Zn-free samples and in Table 4 for the samples in equilibrium with Cu and Fe alloys. Thermodynamic foundation of this principle is the fact that the proportion of phases does not affect the final composition of those phases, as long as the bulk composition remains inside the tie-line. In the case of multicomponent multi-phase equilibria tie-lines become tie-simplices in multi-dimensional space determined by concentration of all elements, as schematically illustrated in Fig. 7. For this type of calculations, cumulative differences in the compositions of phases, predicted by the model and measured in experiment, will result in a shift of predicted proportions from initial target proportions. Example of #1 in Table 3: due to the difference in slag and spinel, the predicted proportion of spinel is 15 %, compared to the target of 10 %. Target proportions are given in Table 3, Tables 4 and 5. Thus, the overall accuracy of model predictions can be shown in simpler terms (see Fig. 7).

Results for the samples at fixed $p(O_2)$ are given in Table 3. Predicted composition of the oxide liquid was characterized by the overestimation of CaO and SiO_2, by 0.32 and 2.0 wt%, respectively, and underestimation of "FeO" and $AlO_{1.5}$, by 0.6 and 1.8 wt%, respectively. At the same time the predicted concentrations of "FeO" and $AlO_{1.5}$ in spinel were found to be somewhat shifted from the experiment: "FeO" to the lower, while $AlO_{1.5}$ to higher concentrations, by 0.7 and 0.9 wt%, respectively. These cumulative deviations result in systematic proportion of spinel to be overestimated by 4.5 ± 0.3 mass %.

Samples in equilibrium with liquid copper and solid iron alloys, as well as spinel oxide phase are collected in Table 4. Predicted compositions of oxide liquid were found to be within 1.0 wt% from the experimental measurements for every oxide component (Table 4). The composition of spinel indicated systematic shift in Fe/Zn ratio: ZnO was underestimated by 6.7 ± 1.9 wt%, while "FeO" was overestimated by 7.6 ± 2.4 wt%. The proportion of spinel was overestimated on average by 1.9 ± 0.9 mass %. No systematic deviation was found in the predictions for the Zn in the liquid copper alloy. Fe in liquid copper alloy was found to be overestimated by 2.1 ± 1.9 wt%. The overestimation of concentration of Fe in liquid copper-zinc alloy was found to be consistent with the recent findings by Sineva et al. [67].

The differences between the experimental results and the respective predictions indicated above will form the basis for the subsequent

Table 3
Comparison of the predicted phase compositions and proportions for oxide liquids (slags) in equilibrium with spinel and copper liquid alloy at fixed $p(O_2)$.

#	T, °C	Exp./Pred	Phase	Composition, wt.%					Mass %	Ox./*Me.*
				"$CuO_{0.5}$"	"FeO"	CaO	SiO_2	$AlO_{1.5}$		Ox
				Cu	Fe	Ca	Si	Al		Me
1	1300	Exp.	oxide liquid	1.4	28.8	6.3	38.8	24.7	85.0	Ox
			spinel	0.13	42.3	0.06	0.07	57.4	10.0	Ox
			Cu metal	***99.85***	***0.14***	***0.01***	***0.00***	***0.00***	***5.0***	***Me***
		Pred.	oxide liquid	1.3	27.9	6.8	41.3	22.7	79.9	Ox
			spinel	0.02	43.0	0.00	0.00	56.9	15.0	Ox
			Cu metal	***99.83***	***0.17***	***0.00***	***0.00***	***0.00***	***5.1***	***Me***
2	1300	Exp.	oxide liquid	1.4	25.7	6.6	40.4	25.9	85.0	Ox
			spinel	0.19	42.0	0.08	0.1	57.6	10.0	Ox
			Cu metal	***99.81***	***0.14***	***0.01***	***0.02***	***0.01***	***5.0***	***Me***
		Pred.	oxide liquid	1.4	24.8	6.9	42.7	24.2	80.5	Ox
			spinel	0.02	42.1	0.00	0.00	57.8	14.5	Ox
			Cu metal	***99.87***	***0.13***	***0.00***	***0.00***	***0.00***	***5.1***	***Me***
3	1300	Exp.	oxide liquid	0.72	25.0	6.7	41.0	26.6	85.0	Ox
			spinel	0.04	40.8	0.04	0.04	59.1	10.0	Ox
			Cu metal	***99.43***	***0.43***	***0.01***	***0.03***	***0.00***	***5.0***	***Me***
		Pred.	oxide liquid	0.72	24.0	7.1	43.4	24.8	80.4	Ox
			spinel	0.01	41.0	0.00	0.00	58.8	14.6	Ox
			Cu metal	***99.52***	***0.47***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
4	1300	Exp.	oxide liquid	0.76	27.1	6.5	40.0	25.6	85	Ox
			spinel	0.13	41.6	0.02	0.06	58.2	10.0	Ox
			Cu metal	***99.51***	***0.37***	***0.01***	***0.02***	***0.00***	***5.0***	***Me***
		Pred.	oxide liquid	0.76	26.2	6.9	42.3	23.8	80.4	Ox
			spinel	0.01	41.7	0.00	0.00	58.2	14.5	Ox
			Cu metal	***99.52***	***0.47***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
5	1200	Exp.	oxide liquid	1.5	46.7	6.4	33.4	12.0	85.0	Ox
			spinel	0.05	65.6	0.26	0.21	33.8	10.0	Ox
			Cu metal	***99.95***	***0.04***	***0.00***	***0.01***	***0.00***	***5.0***	***Me***
		Pred.	oxide liquid	1.5	46.3	6.8	35.2	10.3	80.7	Ox
			spinel	0.01	62.6	0.00	0.00	37.4	14.3	Ox
			Cu metal	***99.89***	***0.1***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
6	1200	Exp.	oxide liquid	0.77	53.2	6.9	29.2	9.9	85.0	Ox
			spinel	0.04	57.8	0.08	0.12	41.9	10.0	Ox
			Cu metal	***99.73***	***0.24***	***0.01***	***0.01***	***0.01***	***5.0***	***Me***
		Pred.	oxide liquid	0.76	53.2	7.3	30.7	8.0	80.6	Ox
			spinel	0.00	56.5	0.00	0.00	43.4	14.3	Ox
			Cu metal	***99.54***	***0.46***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
7	1200	Exp.	oxide liquid	0.81	52.9	6.5	29.6	10.2	85.0	Ox
			spinel	0.04	57.9	0.12	0.21	41.7	10.0	Ox
			Cu metal	***99.78***	***0.19***	***0.01***	***0.01***	***0.01***	***5.0***	***Me***
		Pred.	oxide liquid	0.80	52.9	6.9	31.1	8.3	80.7	Ox
			spinel	0.00	56.1	0.00	0.00	43.8	14.3	Ox
			Cu metal	***99.58***	***0.42***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
Δ		oxide liquid	average	0.00	−0.56	0.38	2.0	−1.8	−4.5	–
			stdev	0.00	0.46	0.02	0.41	0.10	0.28	–
			RMS[a]	0.00	0.70	0.38	2.1	1.8	4.5	–
			max	0.00	0.07	0.41	2.5	−1.7	−4.3	–
			min	0.00	−0.95	0.35	1.5	−2.0	−5.1	–
		spinel	average	−0.08	−0.7	−0.09	−0.11	0.94	4.5	–
			stdev	0.06	1.4	0.08	0.07	1.5	0.27	–
			RMS[a]	0.09	1.5	0.12	0.13	1.7	4.5	–
			max	−0.03	0.72	−0.01	−0.04	3.5	5.0	–
			min	−0.17	−3.1	−0.25	−0.21	−0.47	4.3	–
		Cu	***average***	***−0.04***	***0.09***	***−0.01***	***−0.01***	***−0.01***	***0.04***	-
			stdev	***0.12***	***0.09***	***0.00***	***0.01***	***0.00***	***0.02***	-
			RMS[a]	***0.12***	***0.13***	***0.01***	***0.02***	***0.01***	***0.04***	-
			max	***0.09***	***0.23***	***0.00***	***0.00***	***0.00***	***0.07***	-
			min	***−0.21***	***−0.01***	***−0.01***	***−0.03***	***−0.01***	***0.02***	-

[a] Root mean square: $\mathrm{RMS} = \sqrt{\left(\frac{1}{n}\sum_i \Delta x_i^2\right)}$.

reoptimisation of the thermodynamic model parameters of the Cu–Pb–Zn–Fe–Ca–Si–Al–Mg–O–S-(As, Sn, Sb, Bi, Ag, Au, Ni, Cr, Co and Na) gas/oxide liquid/matte/speiss/metal/solids system. E.g. the systematic shift in the Fe/Zn ratio indicated that the parameters in the oxide liquid or spinel solutions (both involve Fe–Zn and can affect the predictions) were chosen incorrectly and need to be modified to comply with the results of the present and available previous studies.

3.3. Additional series of experiments for further database development

Analysis of model uncertainties for the samples containing liquid oxide (slag) in equilibrium with liquid copper, solid iron alloys, and solid oxide phases are presented in Table 5. Any improvements in multicomponent systems should be done maintaining self-consistency in the lower-order systems. Present study is linked to the recent developments in the binary, ternary and quaternary subsystems of the

Table 4
Comparison of the predicted phase compositions and proportions for oxide liquids (slags) in equilibrium with spinel, copper liquid and iron solid alloy.

#	T, °C	Exp./Pred.	Phase	Composition, wt.%						Mass %	Ox./*Me.*
				"$CuO_{0.5}$"	ZnO	"FeO"	CaO	SiO_2	$AlO_{1.5}$		Ox
				Cu	Zn	Fe	*Ca*	Si	Al		*Me*
8	1320	Exp.	oxide liquid	0.30	2.2	49.7	6.1	28.2	13.6	80.0	Ox
			spinel	0.05	9.0	34.6	0.08	0.09	56.1	10.0	Ox
			Cu alloy	***92.7***	***4.0***	***3.3***	***0.01***	***0.00***	***0.00***	***5.0***	***Me***
			Fe alloy	***10.0***	***0.25***	***89.4***	***0.06***	***0.18***	***0.07***	***5.0***	***Me***
		Pred.	oxide liquid	0.37	2.6	48.8	6.2	28.5	13.5	79.2	Ox
			spinel	0.00	5.6	38.6	0.00	0.00	55.8	10.5	Ox
			Cu alloy	***87.6***	***3.4***	***9.0***	***0.00***	***0.00***	***0.00***	***5.2***	***Me***
			Fe alloy	***9.9***	***0.2***	***89.9***	***0.00***	***0.00***	***0.00***	***5.1***	***Me***
9	1220	Exp.	oxide liquid	0.21	2.7	59.4	5.8	24.3	7.6	80.0	Ox
			spinel	0.02	11.6	35.5	0.11	0.18	52.6	10.0	Ox
			Cu alloy	***93.4***	***3.4***	***3.2***	***0.02***	***0.02***	***0.01***	***5.0***	***Me***
			Fe alloy	***9.2***	***0.23***	***90.6***	***0.03***	***0.00***	***0.02***	***5.0***	***Me***
		Pred.	oxide liquid	0.32	3.3	59.0	6.0	25.1	6.2	77.4	Ox
			spinel	0.00	5.8	41.8	0.00	0.00	52.4	12.6	Ox
			Cu alloy	***91.5***	***3.1***	***5.4***	***0.00***	***0.00***	***0.00***	***5.1***	***Me***
			Fe alloy	***8.6***	***0.14***	***91.3***	***0.00***	***0.00***	***0.00***	***4.9***	***Me***
10	1220	Exp.	oxide liquid	0.18	6.1	54.4	5.8	26.4	7.1	80.0	Ox
			spinel	0.02	21.5	24.4	0.10	0.23	53.8	10.0	Ox
			Cu alloy	***88.9***	***7.4***	***3.7***	***0.02***	***0.01***	***0.01***	***5.0***	***Me***
			Fe alloy	***8.2***	***0.43***	***91.3***	***0.02***	***0.01***	***0.01***	***5.0***	***Me***
		Pred.	oxide liquid	0.31	7.3	53.3	5.9	26.9	6.3	78.7	Ox
			spinel	0.00	12.5	33.8	0.00	0.00	53.7	11.3	Ox
			Cu alloy	***88.4***	***6.5***	***5.0***	***0.00***	***0.00***	***0.00***	***4.9***	***Me***
			Fe alloy	***8.1***	***0.32***	***91.6***	***0.00***	***0.00***	***0.00***	***5.1***	***Me***
11	1220	Exp.	oxide liquid	0.20	3.3	54.6	6.2	27.6	8.1	80.0	Ox
			spinel	0.04	14.6	30.9	0.15	0.36	53.9	10.0	Ox
			Cu alloy	***92.3***	***4.4***	***3.3***	***0.00***	***0.01***	***0.02***	***5.0***	***Me***
			Fe alloy	***7.7***	***0.27***	***92.0***	***0.00***	***0.00***	***0.01***	***5.0***	***Me***
		Pred.	oxide liquid	0.30	4.1	53.8	6.4	28.5	6.8	77.5	Ox
			spinel	0.00	7.4	38.8	0.00	0.00	53.7	12.3	Ox
			Cu alloy	***91.5***	***3.6***	***4.9***	***0.00***	***0.00***	***0.00***	***4.9***	***Me***
			Fe alloy	***8.3***	***0.15***	***91.6***	***0.00***	***0.00***	***0.00***	***5.3***	***Me***
12	1170	Exp.	oxide liquid	0.15	3.0	54.1	6.6	28.9	7.3	80.0	Ox
			spinel	0.01	13.6	32.5	0.18	0.42	53.3	10.0	Ox
			Cu alloy	***92.8***	***3.4***	***3.8***	***0.01***	***0.00***	***0.01***	***5.0***	***Me***
			Fe alloy	***7.6***	***0.15***	***92.2***	***0.01***	***0.01***	***0.00***	***5.0***	***Me***
		Pred.	oxide liquid	0.28	3.8	53.4	6.7	29.5	6.3	78.3	Ox
			spinel	0.00	6.7	39.8	0.00	0.00	53.5	11.8	Ox
			Cu alloy	***92.7***	***3.0***	***4.3***	***0.00***	***0.00***	***0.00***	***4.9***	***Me***
			Fe alloy	***7.9***	***0.12***	***92.0***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
13	1120	Exp.	oxide liquid	0.13	1.8	44.6	5.6	36.8	11.1	80.0	Ox
			spinel	0.04	14.2	26.8	0.03	0.14	58.8	10.0	Ox
			Cu alloy	***95.2***	***2.5***	***2.3***	***0.03***	***0.00***	***0.01***	***5.0***	***Me***
			Fe alloy	***7.8***	***0.06***	***92.1***	***0.00***	***0.00***	***0.02***	***5.0***	***Me***
		Pred.	oxide liquid	0.23	2.7	43.3	5.9	38.3	9.7	76.9	Ox
			spinel	0.00	6.3	37.4	0.00	0.00	56.2	13.0	Ox
			Cu alloy	***94.4***	***2.2***	***3.4***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
			Fe alloy	***7.3***	***0.07***	***92.6***	***0.00***	***0.00***	***0.00***	***5.1***	***Me***
Δ		oxide liquid	average	0.11	0.81	−0.87	0.17	0.79	−1.0	−2.0	–
			stdev	0.02	0.23	0.32	0.06	0.41	0.51	0.88	–
			RMS[a]	0.11	0.84	0.91	0.18	0.87	1.1	2.1	–
			max	0.13	1.1	−0.39	0.23	1.5	−0.11	−0.77	–
			min	0.07	0.46	−1.3	0.08	0.33	−1.4	−3.1	–
		spinel	average	−0.03	−6.7	7.6	−0.11	−0.24	−0.54	1.9	–
			stdev	0.01	1.9	2.4	0.05	0.13	1.0	0.94	–
			RMS[a]	0.03	6.9	7.9	0.12	0.27	1.1	2.1	–
			max	−0.01	−3.4	10.7	−0.03	−0.09	0.17	3.0	–
			min	−0.04	−9.0	4.0	−0.18	−0.42	−2.6	0.46	–
		Cu alloy	***average***	***−1.5***	***−0.56***	***2.1***	***−0.01***	***−0.01***	***−0.01***	***0.00***	-
			stdev	***1.9***	***0.24***	***1.9***	***0.01***	***0.01***	***0.01***	***0.14***	-
			RMS[a]	***2.3***	***0.60***	***2.7***	***0.02***	***0.01***	***0.01***	***0.13***	-
			max	***−0.03***	***−0.32***	***5.7***	***0.00***	***0.00***	***0.00***	***0.25***	-
			min	***−5.2***	***−0.92***	***0.49***	***−0.03***	***−0.02***	***−0.02***	***−0.12***	-
		Fe alloy	***average***	***−0.08***	***−0.06***	***0.22***	***−0.02***	***−0.03***	***−0.02***	***0.06***	-
			stdev	***0.43***	***0.05***	***0.44***	***0.02***	***0.07***	***0.03***	***0.11***	-
			RMS[a]	***0.40***	***0.08***	***0.46***	***0.03***	***0.07***	***0.03***	***0.12***	-
			max	***0.54***	***0.01***	***0.70***	***0.00***	***0.00***	***0.00***	***0.25***	-
			min	***−0.56***	***−0.12***	***−0.42***	***−0.06***	***−0.18***	***−0.07***	***−0.08***	-

[a] Root mean square: $\mathrm{RMS} = \sqrt{\left(\frac{1}{n}\sum_i \Delta x_i^2\right)}$.

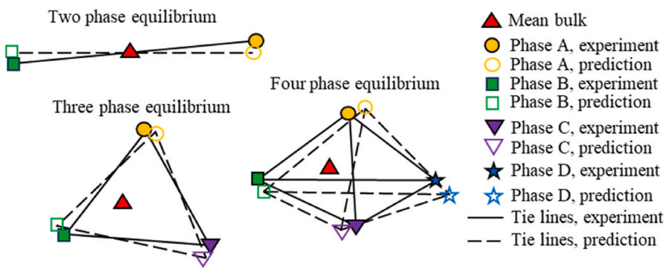

Fig. 7. Schematic demonstration of using "mean bulk" method to compare experimental and thermodynamic prediction for two, three and four-phase equilibria in multi-component system.

system of interest [27,60,68–71]. Points 17–19 in Table 5 target phase equilibria among oxide liquid, tridymite and willemite in the ZnO–FeO–$FeO_{1.5}$-SiO_2 system with small addition of $AlO_{1.5}$ and CaO. The experimental results were then compared with the model predictions in the similar way to the samples of industrial interest (Table 5). The results indicated small, but important signals for further improvement: underestimation of ZnO and "FeO" and overestimation of CaO and $AlO_{1.5}$ in feldspar and overestimation of spinel stability (samples #16 and 17). The predicted solubility of "$CuO_{0.5}$" in oxide liquid was found to be in agreement with experiments.

The outlined comparison, together with that described for the samples of industrial interest, is to be used in the continuous development of the thermodynamic model of the Cu–Pb–Zn–Fe–Ca–Si–Al–Mg–O–S-(As,

Table 5
Comparison of the predicted phase compositions and proportions for oxide liquids (slags) in equilibrium with tridymite, willemite, feldspar, copper liquid alloy and iron solid alloy.

#	T, °C	Exp./Pred.	Phase	Composition, wt.%						Mass %	Ox./*Me.*
				"$CuO_{0.5}$"	ZnO	"FeO"	CaO	SiO_2	$AlO_{1.5}$		Ox
				Cu	Zn	Fe	Ca	Si	Al		Me
14	1150	Exp.	liquid	0.12	21.6	18.9	5.3	44.7	9.5	75.0	Ox
			spinel	0.08	39.6	4.1	0.10	0.44	55.7	5.0	Ox
			tridymite[a]	0.03	0.96	0.6	0.11	98.1	0.19	5.0	Ox
			willemite	0.07	57.7	15.1	0.04	27.0	0.11	5.0	Ox
			Cu alloy	***70.2***	***27.7***	***2.0***	***0.01***	***0.00***	***0.01***	***5.0***	***Me***
			Fe alloy	***4.9***	***1.4***	***93.6***	***0.02***	***0.00***	***0.02***	***5.0***	***Me***
		Pred.	liquid	0.14	18.4	19.4	6.0	46.6	9.4	65.7	Ox
			spinel	0.00	36.4	7.6	0.00	0.00	56.0	6.6	Ox
			tridymite	0.00	0.00	0.00	0.00	100.0	0.00	5.7	Ox
			willemite	0.00	55.0	17.4	0.00	27.6	0.00	12.1	Ox
			Cu alloy	***72.3***	***24.7***	***3.0***	***0.00***	***0.00***	***0.00***	***4.9***	***Me***
			Fe alloy	***4.8***	***3.4***	***91.8***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
15	1100	Exp.	liquid	0.09	17.8	14.9	10.0	47.5	9.8	75.0	Ox
			feldspar[a]	0.09	4.3	1.2	17.8	44.0	32.5	5.0	Ox
			tridymite[a]	0.12	0.94	0.43	0.29	97.9	0.27	5.0	Ox
			willemite	0.05	59.1	13.0	0.25	27.5	0.08	5.0	Ox
			Cu alloy	***74.6***	***23.0***	***2.2***	***0.17***	***0.00***	***0.00***	***5.0***	***Me***
			Fe alloy	***4.3***	***1.5***	***94.1***	***0.10***	***0.00***	***0.00***	***5.0***	***Me***
		Pred.	liquid	0.09	15.8	15.5	10.5	48.8	9.4	64.3	Ox
			feldspar	0.00	0.57	0.00	19.7	44.5	35.2	8.5	Ox
			tridymite	0.00	0.00	0.00	0.00	100.0	0.00	5.6	Ox
			willemite	0.00	55.7	16.8	0.00	27.6	0.00	11.6	Ox
			Cu alloy	***75.1***	***22.5***	***2.4***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
			Fe alloy	***4.7***	***1.4***	***93.9***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
16	1090	Exp.	liquid	0.09	14.8	19.9	8.8	46.9	9.5	75.0	Ox
			feldspar[a]	0.23	1.9	1.5	18.9	45.0	32.5	5.0	Ox
			tridymite[a]	0.00	0.09	0.53	0.17	99.01	0.2	5.0	Ox
			willemite	0.00	53.5	17.6	0.15	28.6	0.09	5.0	Ox
			Cu alloy	***81.7***	***16.4***	***1.8***	***0.03***	***0.01***	***0.01***	***5.0***	***Me***
			Fe alloy	***4.9***	***0.87***	***94.2***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
		Pred.	liquid	0.11	14.6	18.8	9.2	47.8	9.5	80.8	Ox
			feldspar	0.00	0.42	0.00	19.8	44.4	35.4	0.76	Ox
			spinel	0.00	34.6	9.3	0.00	0.00	56.0	1.5	Ox
			tridymite	0.00	0.00	0.00	0.00	100.0	0.00	4.1	Ox
			willemite	0.00	52.8	19.6	0.00	27.7	0.00	2.6	Ox
			Cu alloy	***79.2***	***18.4***	***2.4***	***0.00***	***0.00***	***0.00***	***5.1***	***Me***
			Fe alloy	***5.1***	***0.96***	***94.0***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
17	1115	Exp.	liquid	0.06	13.5	19.0	8.8	48.0	10.6	80.0	Ox
			feldspar[a]	0.08	1.9	1.2	18.5	44.8	33.5	5.0	Ox
			tridymite[a]	0.06	0.12	0.39	0.16	99.04	0.23	5.0	Ox
			Cu alloy	***78.7***	***18.5***	***2.4***	***0.32***	***0.00***	***0.02***	***5.0***	***Me***
			Fe alloy	***6.2***	***1.0***	***92.7***	***0.00***	***0.01***	***0.00***	***5.0***	***Me***
		Pred.	liquid	0.12	12.9	18.3	9.0	49.0	10.6	82.8	Ox
			feldspar	0.00	0.3	0.00	19.7	44.5	35.5	2.9	Ox
			spinel	0.00	33.8	10.1	0.00	0.00	56.1	0.60	Ox
			tridymite	0.00	0.00	0.00	0.00	100.0	0.00	3.8	Ox
			Cu alloy	***78.8***	***18.5***	***2.7***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
			Fe alloy	***5.3***	***1.0***	***93.7***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
18	1200	Exp.	liquid	0.11	22.2	39.6	0.02	38.0	0.10	80.0	Ox
			tridymite[a]	0.03	0.52	0.63	0.02	98.8	0.01	5.0	Ox
			willemite	0.06	47.5	24.3	0.01	28.1	0.00	5.0	Ox
			Cu alloy	***78.3***	***20.2***	***1.4***	***0.02***	***0.01***	***0.00***	***5.0***	***Me***
			Fe alloy	***6.2***	***1.2***	***92.6***	***0.01***	***0.01***	***0.00***	***5.0***	***Me***

(*continued on next page*)

Table 5 (*continued*)

#	T, °C	Exp./Pred.	Phase	Composition, wt.%						Mass %	Ox./*Me.*
				"$CuO_{0.5}$"	ZnO	"FeO"	CaO	SiO_2	$AlO_{1.5}$		Ox
				Cu	Zn	Fe	Ca	Si	Al		Me
		Pred.	liquid	0.18	20.2	42.4	0.05	37.1	0.12	68.5	Ox
			tridymite	0.00	0.00	0.00	0.00	100.0	0.00	7.3	Ox
			willemite	0.00	45.9	26.2	0.00	27.9	0.00	14.4	Ox
			Cu alloy	***79.3***	***16.7***	***3.9***	***0.00***	***0.00***	***0.00***	***4.9***	***Me***
			Fe alloy	***6.4***	***1.1***	***92.5***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
19	1200	Exp.	liquid	0.10	23.5	28.2	6.3	41.8	0.07	80.0	Ox
			tridymite[a]	0.03	0.52	0.63	0.02	98.8	0.01	5.0	Ox
			willemite	0.00	54.0	18.2	0.13	27.7	0.00	5.0	Ox
			Cu alloy	***72.9***	***24.1***	***2.9***	***0.03***	***0.01***	***0.00***	***5.0***	***Me***
			Fe alloy	***5.1***	***1.5***	***93.4***	***0.01***	***0.01***	***0.00***	***5.0***	***Me***
		Pred.	liquid	0.12	23.6	28.3	6.4	41.6	0.08	79.3	Ox
			tridymite	0.00	0.00	0.00	0.00	100.0	0.00	5.3	Ox
			willemite	0.00	51.3	21.0	0.00	27.7	0.00	5.6	Ox
			Cu alloy	***73.6***	***22.6***	***3.8***	***0.00***	***0.00***	***0.00***	***4.9***	***Me***
			Fe alloy	***5.5***	***3.2***	***91.3***	***0.00***	***0.00***	***0.00***	***5.0***	***Me***
Δ		liquid	average	0.03	−1.3	0.36	0.32	0.67	−0.08	−3.9	–
			stdev	0.03	1.3	1.4	0.28	1.1	0.19	7.5	–
			RMS[b]	0.04	1.7	1.3	0.41	1.2	0.19	7.9	–
			max	0.06	0.07	2.8	0.77	1.9	0.04	5.8	–
			min	0.00	−3.1	−1.1	0.03	−0.92	−0.45	−11.5	–
		feldspar	average	−0.15	−1.5	−1.3	1.0	−0.46	2.4	−3.2	–
			stdev	0.11	0.09	0.18	0.19	0.24	0.62	1.5	–
			RMS[b]	0.17	1.5	1.3	1.1	0.49	2.5	3.4	–
			max	−0.08	−1.5	−1.2	1.2	−0.28	2.9	−2.1	–
			min	−0.23	−1.6	−1.5	0.91	−0.63	2.0	−4.2	–
		spinel	average	−0.09	−3.5	1.2	0.89	0.06	1.5	2.6	–
			stdev	0.01	0.39	3.4	1.4	0.70	1.7	1.3	–
			RMS[b]	0.09	3.5	2.7	1.3	0.50	1.9	2.7	–
			max	−0.08	−3.2	3.5	1.9	0.55	2.6	3.5	–
			min	−0.09	−3.8	−1.2	−0.1	−0.44	0.29	1.6	–
		willemite	average	−0.04	−2.3	2.6	−0.12	−0.09	−0.06	4.2	–
			stdev	0.03	1.0	0.78	0.09	0.54	0.05	4.9	–
			RMS[b]	0.05	2.4	2.7	0.14	0.49	0.07	6.1	–
			max	0.00	−0.77	3.8	−0.01	0.58	0.00	9.4	–
			min	−0.07	−3.4	1.9	−0.25	−0.91	−0.11	−2.4	–
		Cu alloy	***average***	***0.32***	***−1.1***	***0.9***	***−0.10***	***−0.01***	***−0.01***	***−0.04***	**-**
			stdev	***1.5***	***2.0***	***0.85***	***0.12***	***0.01***	***0.01***	***0.10***	**-**
			RMS[b]	***1.4***	***2.2***	***1.2***	***0.15***	***0.01***	***0.01***	***0.10***	**-**
			max	***2.1***	***2.0***	***2.5***	***−0.01***	***0.00***	***0.00***	***0.14***	**-**
			min	***−2.5***	***−3.5***	***0.25***	***−0.32***	***−0.01***	***−0.02***	***−0.12***	**-**
		Fe alloy	***average***	***0.01***	***0.6***	***−0.58***	***−0.02***	***0.00***	***0.00***	***0.00***	**-**
			stdev	***0.49***	***0.99***	***1.2***	***0.04***	***0.00***	***0.01***	***0.00***	**-**
			RMS[b]	***0.45***	***1.1***	***1.2***	***0.04***	***0.01***	***0.01***	***0.00***	**-**
			max	***0.39***	***2.0***	***0.93***	***0.00***	***0.00***	***0.00***	***0.01***	**-**
			min	***−0.92***	***−0.14***	***−2.1***	***−0.10***	***−0.01***	***−0.02***	***0.00***	**-**

[a] the measurement of the phase composition was affected by the close presence of other phases.

[b] Root mean square: $RMS = \sqrt{\left(\frac{1}{n}\sum_i \Delta x_i^2\right)}$.

Sn, Sb, Bi, Ag, Au, Ni, Cr, Co and Na) gas/oxide liquid/matte/speiss/metal/solids system to improve the accuracies of the predictions of phase equilibria in the high-temperature pyrometallurgical primary and recycling processes.

4. Conclusions

Experimental data on the slag chemistry within the "$CuO_{0.5}$"-ZnO-FeO-$FeO_{1.5}$-CaO-SiO_2-$AlO_{1.5}$ system for conditions important for WEEE smelting using black copper route have been obtained for the first time. Concentration of copper in liquid oxide (slag) has been measured as a function of oxygen partial pressure over the system. The experimental results were combined with thermodynamic calculations to demonstrate the oxide primary phase fields existing in the area of interest. These data are valuable to operate the reduction smelting furnace at optimal proportion of solids in slag.

A computational method was developed to test the accuracy of predictions of thermodynamic model. Thermodynamic model developed earlier based on binary, ternary and quaternary subsystems demonstrated reasonable performance in describing spinel liquidus and proportion of solids in this 7-component system. Systematic deviations of model predictions from experimental data have been quantified, and improvements are planned.

CRediT authorship contribution statement

Georgii Khartcyzov: Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Cora Kleeberg:** Conceptualization, Investigation, Validation, Writing – original draft, Writing – review & editing. **Maksym Shevchenko:** Conceptualization, Investigation, Methodology, Supervision, Writing – original draft, Writing – review & editing. **Denis Shishin:** Conceptualization, Investigation, Methodology, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Evgueni Jak:** Conceptualization, Investigation, Methodology, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank Australian Research Council, linkage project LP190101020 “Future copper metallurgy for the age of e-mobility and the circular economy”, and industry partners: Anglo American Pt (South Africa), Aurubis AG (Germany), BHP Olympic Dam (Australia), Boliden (Sweden), Glencore Technology (Australia), Kazzinc (Kazakhstan), PASAR (Philippines), Metso (Finland), Peñoles (Mexico), RHI Magnesita (Austria), Rio Tinto Kennecott (USA), Umicore NV (Belgium) for providing funding and technical support. Additionally, authors would like to thank staff of the University of Queensland Centre for Microanalysis and Microscopy (CMM) for their support in maintenance and operation of scanning electron microprobe and XRD facilities in the Centre and Prof. Peter Charles Hayes for his valuable contributions to the research done at the Pyrometallurgy Innovation Laboratory.

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